Lecture 8

Quantum non-locality and entanglement

What is the difference between a shared uniformly random bit

$$\frac{1}{2}[00] + \frac{1}{2}[11] = \frac{1}{2}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$$

and a shared maximally entangled state

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix} ?$$

Clearly, the quantum state is at least as useful since it reduces to a shared uniformly random bit when both qubits are measured in the standard basis. But can it be more useful?

Shared random bits represent correlations. When Alice and Bob look at their bits, the values are correlated – in case of the uniformly random bit, if Alice sees 0, she immediately knows that Bob's value will also be 0, even if he is at the other end of the galaxy. Even if Alice now knows Bob's value, this does not help Bob in any way – he does not know if Alice has looked at her bit and already knows what value he will see. For all he knows, he might still be the first one looking at the bit and hence might in principle still observe either of the two values. Note that there is no inconsistency here – in whichever order they observe the bits, they will see 00 w.p. $\frac{1}{2}$ and 11 w.p. $\frac{1}{2}$. Also, this does not let them communicate instantaneously in any way.

Quantum correlations

Quantumly the situation is very similar, except each qubit can be measured in an arbitrary basis. But how are these measurement outcomes correlated?

Ex Consider $|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, called the EPR (Einstein-Podolsky-Rosen) state. You can check that

$$(U \otimes U^*)|\beta_{00}\rangle = |\beta_{00}\rangle$$

for any unitary U, meaning that measurement outcomes are perfectly correlated if the first qubit is measured in the U basis and the second qubit is measured in the U^* basis, for any single-qubit unitary U.

Ex Consider $|\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, also called the singlet or antisymmetric state. You can check that

$$(U \otimes U)|\beta_{11}\rangle = \det(U)|\beta_{11}\rangle$$

($\det(U)$: a global phase when U is unitary)

for any 2×2 matrix U. Hence, if both qubits are measured in the U basis, the measurement outcomes are perfectly anti-correlated (opposite and uniformly random), for any unitary U.

These examples illustrate a sense in which entanglement represents a stronger (quantum) correlation because the measurement outcomes of both qubits are perfectly related to each other not only in the standard basis but in any matching pair of bases.

Just like with probabilistic bits, it is not possible to use this phenomenon to communicate between the two parties.

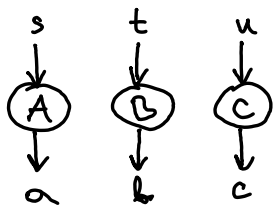
Quantum non-locality

Neither quantum nor classical correlations let you communicate. But does that also mean that we can reproduce the quantum correlations classically without communication? Surprisingly, the answer is "no". I.e., quantum correlations are strictly more powerful that classical ones and can even achieve tasks that cannot be accomplished classically without communication. This phenomenon is called quantum non-locality — quantum behaviour cannot be reproduced classically by local actions only.

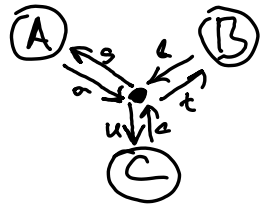
This is illustrated best in the setting of non-local games, where several players collaboratively play against a referee. In the classical case, their strategy can rely on a shared random string, while in the quantum case they can use a shared entangled state.

GHZ game (Greenberger–Horne–Zeilinger)

This game involves three colaborating players: Alice, Bob, and Charlie. Each of them receives a bit, $s,t,u \in \{0,1\}$, and must output a bit, $a,b,c \in \{0,1\}$. They can share a random string or a quantum state before the game starts but cannot



communicate once the game starts. This can be enforced by putting them sufficiently far from each other and requiring the answers to be returned fast enough so that no messages can be exchanged between the parties during the time they need to come up with answers, even if they communicate at the speed of light.



The rules of the game are as follows:

Promise: $s \oplus t \oplus u = 0$ (the input has even parity)

Goal: $a \oplus b \oplus c = \begin{cases} 0 & \text{if } stu = 000 \\ 1 & \text{otherwise, i.e., } stu \in \{011, 101, 110\}. \end{cases}$

E.g., if $stu = 000$, they can output any string $abc \in \{000, 011, 101, 110\}$. I.e., there is more than one correct answer.

Deterministic strategies

Let's assume each party's answer depends deterministically on their input. If Alice receives input $s \in \{0,1\}$, she outputs $a_s \in \{0,1\}$. Her strategy is thus fully specified by two bits, a_0 and a_1. Similarly, Bob's strategy is determined by $b_t \in \{0,1\}$ and Charlie's by $c_u \in \{0,1\}$. To win, their output $a_s b_t c_u$ must satisfy

$$\begin{cases} a_0 \oplus b_0 \oplus c_0 = 0 \\ a_0 \oplus b_1 \oplus c_1 = 1 \\ a_1 \oplus b_0 \oplus c_1 = 1 \\ a_1 \oplus b_1 \oplus c_0 = 1 \end{cases}$$

But this is impossible since adding all equations mod 2 yields $0 = 1$.

Hence, there is no perfect deterministic strategy. The best we can do is to satisfy (any) three out of the four equations. E.g., if $a_i = b_i = 0$ and $c_i = 1$, irrespective of the input, the last three equations are satisfied but the first one is not. What about more general classical strategies that use randomness?

Probabilistic strategies

Alice, Bob, and Charlie could share some random string which they could use to coordinate their actions (part of it could also be used exclusively by, say, Alice, so she can have a local randomized strategy of her own). Let us denote this string by r and its probability of occurring by q_r. There are 4 possible input strings stu, and let's assume each occurs with probability $1/4$. Then the overall probability of success is

$$\sum_r q_r \frac{1}{4} \sum_{stu \in \{000,011,101,110\}} P(s,t,u,r)$$

where $P(s,t,u,r) = 1$ if the deterministic protocol with randomness r succeeds on input stu, and $P(s,t,u,r) = 0$ if it fails. Since $q_r \geq 0$ and $\sum_r q_r = 1$, this probability is upper bounded by

$$\max_r \frac{1}{4} \sum_{stu \in \{000,011,101,110\}} P(s,t,u,r)$$

This is just $\frac{1}{4}$ times the number of instances on which the best deterministic strategy can succeed. As we saw earlier, the best deterministic strategy can succeed on at most 3 out of 4 inputs. Hence the best probabilistic strategy can succeed with probability at most $3/4$.

Quantum strategies

Instead of a random string, Alice, Bob and Charlie could also share a quantum state before the game starts. In this particular case the following state is very useful:

$$|\Psi\rangle = \frac{1}{2}\big(|000\rangle - |011\rangle - |101\rangle - |110\rangle\big).$$

This state can be obtained from the so-called GHZ state: $|\Psi\rangle = U^{\otimes 3}|GHZ\rangle$ where

$$|GHZ\rangle = \frac{1}{\sqrt{2}}\big(|000\rangle + |111\rangle\big) \quad \text{and} \quad U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}.$$

The main advantage for sharing a quantum state instead of a probability distribution is that qubits can be measured in more than one basis. In particular, the basis can be chosen adaptively based on the player's input. This allows them to produce measurement outcomes that are correlated in ways that are not possible classically.

The strategy for using $|\Psi\rangle$ is as follows: each party measures in a basis determined by their input – the standard basis when 0 and the Hadamard basis when 1. More explicitly:

- Alice applies H^s and measures,
- Bob applies H^t and measures,
- Charlie applies H^u and measures.

Hence you can think of a quantum strategy as a function from input bits to a measurement basis, while classically it was a function from the input bits to output bits.

There are four possible inputs:

000
011
101
110

In the first case they just measure $|\Psi\rangle$ in the standard basis. Since $|\Psi\rangle$ is a superposition of even-weight strings, they will get one of them at random, so their output abc will clearly have the property that $a \oplus b \oplus c = 0$, as desired. The other three cases are symmetric, so let's just consider 011. First, note that

$$|\Psi\rangle = \frac{1}{2}\Big(|0\rangle(|00\rangle - |11\rangle) - |1\rangle(|01\rangle + |10\rangle)\Big)$$

$$= \frac{1}{\sqrt{2}}\Big(|0\rangle|\beta_{10}\rangle - |1\rangle|\beta_{01}\rangle\Big),$$

where $|\beta_{10}\rangle$ and $|\beta_{01}\rangle$ are two of the Bell states. In the exercise class you'll show that

$$(H \otimes H)\,|\beta_{zx}\rangle = (-1)^{zx}\,|\beta_{xz}\rangle,$$

so after $I \otimes H \otimes H$ (for input 011) the state becomes

$$\frac{1}{\sqrt{2}}\Big(|0\rangle|\beta_{01}\rangle - |1\rangle|\beta_{10}\rangle\Big)$$

$$= \frac{1}{2}\Big(|0\rangle(|01\rangle + |10\rangle) - |1\rangle(|00\rangle - |11\rangle)\Big)$$

$$= \frac{1}{2}\Big(|001\rangle + |010\rangle - |100\rangle + |111\rangle\Big).$$

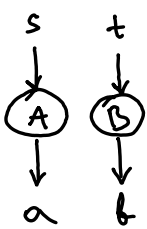
This is a superposition of odd-weight bit strings, so the measurement outcomes a, b, c will satisfy $a \oplus b \oplus c = 1$. Similarly for the other three cases.

Hence the quantum strategy will always succeed and they win with 100% probability.

CHSH game (Clauser-Horne-Shimony-Holt)

The GHZ game is for 3 players. Is there also a two-player game that demonstrates the advantage of quantum strategy?

The CHSH game involves only two players: Alice & Bob.



$s,t \in \{0,1\}$ They win if $a \oplus b = AND(s,t) = st$.

$a,b \in \{0,1\}$

Equivalently:

s t	Win if...
0 0	$a = b$
0 1	$a = b$
1 0	$a = b$
1 1	$a \neq b$

As before, there is no unique winning output in each case, rather the output must satisfy a certain property (namely, have the right parity), and several outputs have this property.

Deterministic strategies

Let a_s denote the bit Alice outputs when she receives s, and b_t denote the bit Bob outputs when he receives t. The winning constraint is then as follows:

$$\begin{cases} a_0 \oplus b_0 = 0 \\ a_0 \oplus b_1 = 0 \\ a_1 \oplus b_0 = 0 \end{cases} \text{ both outputs are the same}$$

$$a_1 \oplus b_1 = 1 \leftarrow \text{their outputs are different}$$

As before, this linear system of equations does not have a solution, since adding all equations mod 2 yields $0 = 1$. One can easily satisfy 3 out of 4 equations.

Probabilistic strategy

By a similar convexity argument, the best success probability under a uniform input distribution is the largest fraction of inputs on which a deterministic strategy can win. As in the GHZ game, this is 3/4 (e.g., both players can always output 0, which satisfies the first three equations).

Quantum strategy

Assume Alice and Bob share the Bell state

$$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle).$$

They will both adjust their basis of measurement depending on their respective inputs. Let

$$G(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$$

Both strategies involve rotating by an angle that depends on the respective input:

- Alice applies $G(\theta_s)$ and measures,
- Bob applies $G(\theta_t)$ and measures,

where $\theta_i = \frac{\pi}{16}(4i-1)$. That is, $\theta_0 = -\pi/16$ and $\theta_1 = 3\pi/16$.

You will check in the exercise class that

$$\big(G(\alpha_1)\otimes G(\alpha_2)\big)|\beta_{10}\rangle = \cos(\alpha_1+\alpha_2)|\beta_{10}\rangle \leftarrow \text{even parity} + \sin(\alpha_1+\alpha_2)|\beta_{01}\rangle, \leftarrow \text{odd parity}$$

where $|\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. When this is measured, we get $a \oplus b = \mathrm{AND}(s,t) = st$ with probability $\cos^2(\pi/8) \approx 0.85 > 3/4$, which is better than classically.

Observables

It is often convenient to use the mathematical formalism of observables to describe the strategies in a non-local game. An *observable* is simply a Hermitian matrix, i.e., a matrix P such that $P^\dagger = P$. Let us consider the spectral decomposition

$$P = \sum_{i=1}^{K} \lambda_i \Pi_i$$

where $\lambda_i \in \mathbb{R}$ are the eigenvalues of P and Π_i are the projectors on the corresponding eigenspaces.

Measuring an observable P simply corresponds to performing the orthogonal measurement with operators $\{\Pi_1, \ldots, \Pi_K\}$ and outputting λ_i instead of i as the measurement outcome.

In case of non-local games, the eigenvalues of the observable being measured are typically just ± 1 so that the output bit $a \in \{0,1\}$ corresponds to the eigenvalue $(-1)^a$. Moreover, if Alice measures an observable A and Bob measures B, their joint measurement is described by $A \otimes B$. This observable has eigenvalues $(-1)^a \cdot (-1)^b = (-1)^{a \oplus b}$ that correspond to the XOR of the two bits.

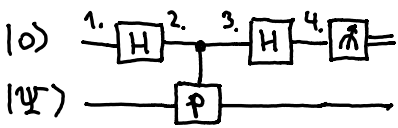
Ex. Measuring in the computational basis corresponds to measuring the observable $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ while measuring in the Hadamard basis corresponds to measuring $|+\rangle\langle +| - |-\rangle\langle -| = HZH = X$. Hence each player's GHZ game strategy is to measure Z if their input is 0 and otherwise measure X.

If a measurement is required to produce several outcomes, one can simultaneously measure several observables. However, for this measurement to be unambiguous, the observables should be simultaneously diagonalizable, i.e., there should be a basis in which all of the observables being measured are diagonal. Then the joint measurement can still be interpreted as a single orthogonal measurement. To determine whether P and Q can be simultaneously measured, one simply needs to check that they commute: $PQ = QP$.

Ex $P = X \otimes X$ and $Q = Z \otimes Z$ can be measured simultaneously, while $P = X$ and $Q = Z$ cannot.

Quantum circuit for measuring an observable

An observable P with eigenvalues ± 1 can be measured on a state $|\Psi\rangle$ using the following circuit that requires an extra qubit initialized in state $|0\rangle$:



1. $|0\rangle|\Psi\rangle$
2. $|+\rangle|\Psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)|\Psi\rangle$
3. $\frac{1}{\sqrt{2}}(|0\rangle|\Psi\rangle + |1\rangle P|\Psi\rangle)$
4. $\frac{1}{2}((|0\rangle+|1\rangle)|\Psi\rangle + (|0\rangle-|1\rangle)P|\Psi\rangle)$
   $= |0\rangle\otimes\left(\frac{I+P}{2}|\Psi\rangle\right) + |1\rangle\otimes\left(\frac{I-P}{2}|\Psi\rangle\right)$

Since P has eigenvalues ± 1, the corresponding two eigenvalues of $\frac{I \pm P}{2}$ are $\frac{1+1}{2} = 1$ and $\frac{1-1}{2} = 0$, respectively, so they are projectors on the two eigenspaces of P and the spectral decomposition of P is given by $P = \left(\frac{I+P}{2}\right) - \left(\frac{I-P}{2}\right)$. Measuring the first qubit projects the second qubit onto one of the two eigenspaces of P, which is equivalent to measuring P.